Department of Computer and Information Systems Engineering

CS-406 Computer Engineering Project

Proposal for the Final Year Design Project

Title	**PhysioGuide - Immersive Physical Rehabilitation System**

Domain	Domain1	Domain2	Domain3	Domain4	Domain5	Domain6
	Virtual Reality (VR) Applications	Health Informatics / Rehabilitation Technology	Machine Learning			

1. Nature of Project[Tick all that applicable]

☑ New Project OR ☐Extension of Existing Project	☑ Industrial Collaboration	☐ Funded
☐ Other Department Collaboration (If yes)Department Name_____	☐ Other Academic Institution Collaboration (If yes)Institution Name_____	

2. Brief Outline(*Problem Identification and Significance*)

Physical rehabilitation requires correct and consistent execution of exercises, which typically need professional supervision. However, therapist availability, distance, and patient motivation are major challenges. This project proposes a VR-based system using OpenPose for pose tracking to guide patients through exercises and provide real-time feedback, allowing for accurate self-rehabilitation and remote therapist monitoring. The system integrates immersive Virtual Reality using the Oculus Quest 2 headset along with external webcams or depth sensors to capture body movements. On the software side, Unity3D (LTS) is used for building the VR environment, while OpenPose, Python, and OpenCV handle pose estimation and motion analysis, with C# scripts managing real-time interaction. The solution leverages datasets for training and evaluation, ensuring accurate detection and feedback across different body types and movement variations. Human involvement includes a team of developers, test users for iterative validation, and advisor feedback to refine the system for usability, accuracy, and medical relevance. Ultimately, this project aims to make physical therapy more accessible, engaging, and effective through an intelligent VR-based rehabilitation platform.

3. Objectives

- Guide patients through upper-body therapeutic exercises in a VR environment.
- Track body movements using OpenPose for real-time pose estimation.
- Provide immediate feedback through visual cues, audio, and text.
- Enable remote progress tracking and session review for therapists.

4. Scope

- Initial focus on 3–5 upper-body exercises (e.g. shoulder abduction, elbow flexion, wrist extensions, neck).
- Unity3D to create a VR environment with virtual trainer and mirrors.
- OpenPose for skeletal key point detection.
- OculusQuest2 to deliver an immersive, wireless VR experience.
- Feedback through color codes(green/red), audio alerts, and on-screen prompts.

5. Proposed Methodology

- **Research and Consultation**: Conduct detailed literature reviews on VR-based rehabilitation and OpenPose use in physical therapy. Interview physical therapists to understand correct exercise form, error types, and real-world use cases. Identify common upper-body therapy movements suitable for VR simulation and real-time pose correction.
- **System Architecture Design**: Design modular system architecture integrating:

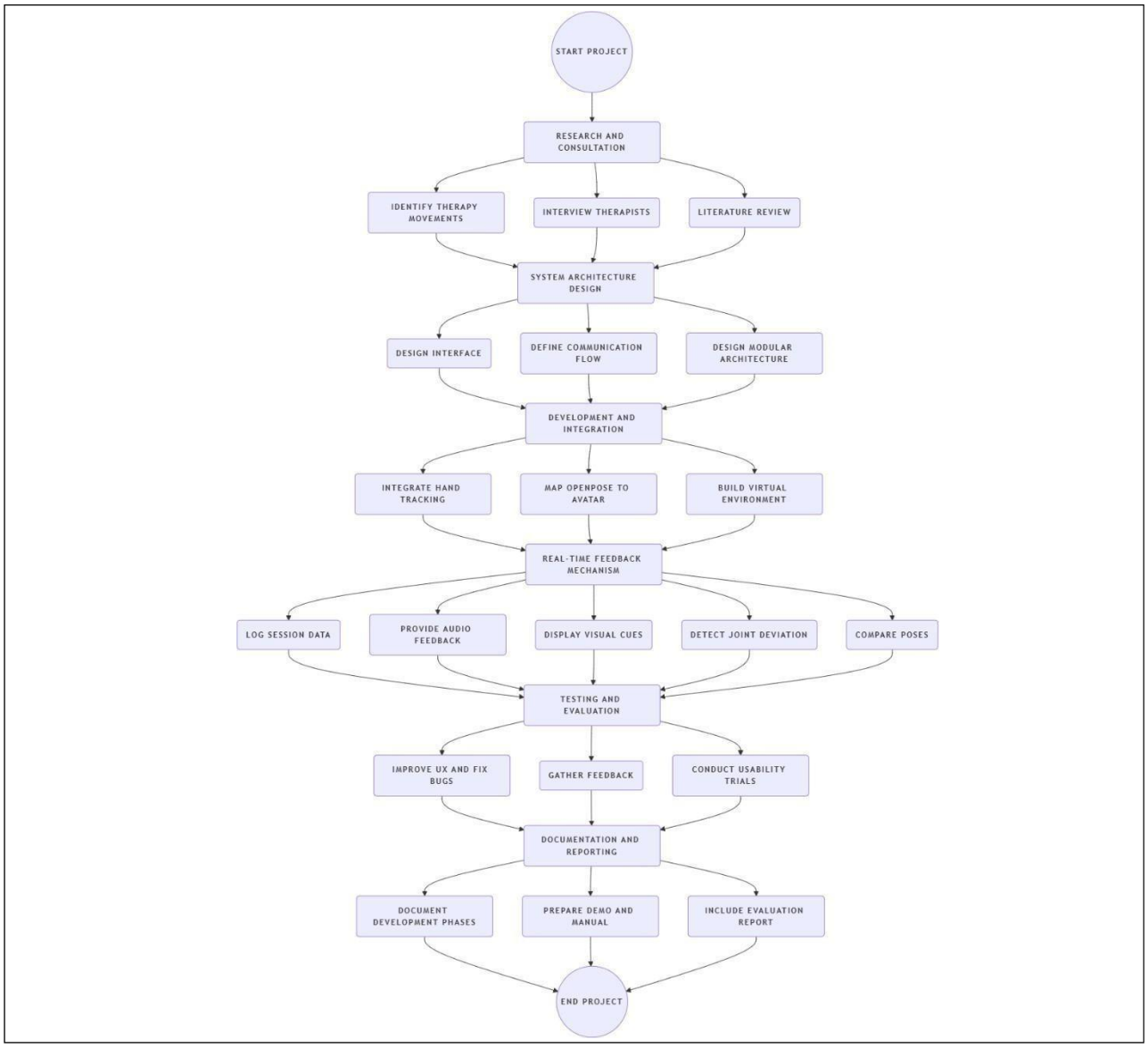
    1. Unity for 3Dimmersive environment.
    2. OpenPose for skeletal track ingvia webcam
    3. OculusQuest2 for VR interaction

  Define communication flow between tracking module and feedback module (using middleware like Python-Unity bridge).Design an intuitive interface for therapists and patients.

- **Development and Integration**: Build a virtual training environment (e.g., VR gym) with realistic assets and mirrored visuals. MapOpenPose skeletal keypoints to a 3D human avatar in Unity, ensuring real-time synchronization for exercises like squats or shoulder raises. Integrate Oculus hand tracking or VR controllers for enhanced motion tracking.
- **Feedback System:** Implement an intelligent feedback system that compares patient poses against reference models. Detects the joint angles and deviation thresholds to trigger real-time visual cues and provide audio guidance. The system will also log session data for progress tracking.
- **Testing & Evaluation**: Conduct usability trials with students or volunteers simulating therapy. Run A/B testing with patients to compare guided vs. unguided sessions, measuring accuracy, usability, and recovery outcomes. Gather this qualitative feedback and improve the UX/UI and correct bugs based on feedback.
- **Documentation & Reporting**: Document each development phase with technical explanations. Include setup instructions for clinics, demo and user manual for therapists. API documentation for developers, including an evaluation report with system limitations and future recommendations.

Deliverables:

- **VR Rehabilitation Environment** – Unity-based immersive VR gym with virtual trainer and mirrors.
- **Therapist–Patient Interfaces** – Integrated pose tracking, real-time feedback (visual/audio), and session logging.
- **Documentation & Evaluation** – Setup guide, user manual, API docs, and final demo with evaluation report.



6. Resources Involved

- **Hardware**: OculusQuest2, webcam/depthsensor.
- **Software**: Unity3D(LTS), OpenPose, Python, OpenCV, C#.
- **Human**: Team of developers, test users, and advisor feedback.

7. Description of Industrial Support(If any)

8. SDGs (If Applicable)

☐ No Poverty	☐ Zero Hunger
☑ Good Health and Well-Being	☑ Quality Education
☐ Gender Equality	☐ Clean water and Sanitation
☐ Affordable and Clean Energy	☑ Decent Work and Economic growth
☑ Industry, Innovations and Infrastructure	☑ Reduced Inequalities
☐ Sustainable Cities and Communities	☐ Responsible Consumption and Production
☐ Climate action	☐ Life Below Water
☐ Life on Land	☐ Peace, Justice and Strong Institutions
☐ Partnerships	

9. Gantt Chart

2025-2026										
Months	Task Name	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr
Task 1	Literature Review& Therapist Consultation	✓	✓							
Task 2	Requirement Analysis and Planning	✓	✓							
Task 3	System Design	✓	✓	✓						
Task 4	Environment And Asset Development			✓	✓					
Task 5	OpenPose Integration			✓	✓	✓				
Task 6	Real-Time Feedback System			✓	✓	✓				
Task 7	Therapist Interface			✓	✓	✓				
Task 8	**MVP: Basic VR gym + OpenPose + feedback**					✓				
Task 9	Testing And User Trials					✓	✓	✓		
Task 10	Bug Fixes and Refinement						✓	✓	✓	
Task 11	Documentation And Report								✓	✓
Task 12	Demo And Evaluation									✓

10. Details of Project Team

i. Students

No.	Name	Seat No.	Signature(s)
1	Jaweria Anwer	CS-22057	
2	Syeda Fatima Mehdvi	CS-22065	
3	Hussain Kazmi	CS-22090	
4	Taha Khan	CS-22134	

ii. Supervisors/ Advisors

	Name	Designation& Department	Address& Contact	Signature(s)
Supervisor	Dr Shahab Tehzeeb			
Co-Supervisor (If any)				
Industrial Advisor (If any)				

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Project Serial No.: _____ Dated: _____	Signature Convener Steering Committee	Signature FYP Coordinator

☐ Proposal Approved	☐ Not Approved	☐ Returned for Clarification/Modification
Comments: (if any)		

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(Signature of Chairperson)

Date: _____